

D412

Network Analytics and Troubleshooting

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A technician, who just finished a certification course, gets a ticket that a branch office cannot print. She has asked users what changed and confirmed the printer shows offline. The branch opened five years ago. What is her next step?

- A. Escalate immediately, since all printer issues require a vendor.
- B. Document the resolution.
- C. Establish a theory of probable cause.
- D. Replace the printer.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

A technician fixed a switch port configuration, and the user's PC now connects. What should happen before closing the ticket?

- A. Establish a new theory of probable cause.
- B. Reboot every switch in the building.
- C. Close it at once since the PC connects.
- D. Verify the user has full function and document the cause and fix.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A user in the accounting department, which recently moved to the third floor, says no websites load. The technician sees the laptop's network link light is off. What should the technician check first?

- A. The cable and switch port.
- B. Reinstall the web browser.
- C. The floor's seating chart.
- D. The DNS server settings.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Staff at a law firm, which has a busy front desk, can reach the file server by its IP address but not by the name files.corp.local. Which tool best tests the likely cause?

- A. nslookup (or dig).
- B. traceroute.
- C. A cable tester.
- D. ping to the IP address.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Users can reach a remote site's server by name some days but not others, and the team wants to see which router along the path stops responding. Which tool fits?

- A. ipconfig.
- B. traceroute.
- C. nslookup.
- D. ping, since it shows every hop.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A laptop at a dental office, recently upgraded to a new operating system, shows IP address 169.254.12.7 and cannot reach anything. What is the most likely cause?

- A. The laptop could not reach a DHCP server.
- B. It has a valid static address from the office.
- C. The DNS server is down.
- D. The new operating system always uses 169.254.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A design firm must copy a 2 GB file to a branch over a 100 Mbps link. The link has 20 ms latency and was installed by a local carrier. Ignoring overhead, about how long will the transfer take?

- A. About 20 seconds.
- B. About 160 seconds.
- C. About 50 seconds.
- D. About 20 milliseconds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

A 500 MB backup must finish in 40 seconds. Ignoring overhead, what minimum throughput is needed?

- A.** 20,000 Mbps
- B.** 12.5 Mbps
- C.** 100 Mbps
- D.** 40 Mbps

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A monitoring dashboard shows a core switch at 65% CPU. The network team, which uses a popular monitoring product, is asked if this is a problem. What is needed to answer?

- A.** Nothing; 65% is always a problem.
- B.** A DNS lookup of the switch name.
- C.** The switch's baseline CPU for this time and load.
- D.** The monitoring product's version number.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

One port on a switch in a warehouse near the loading dock shows CRC errors climbing every minute. The device on that port is a label printer bought last year. Users say printing is slow. What is the most likely cause?

- A.** A DNS misconfiguration.
- B.** The printer is too old.
- C.** A routing loop.
- D.** A damaged cable, interference, or a duplex mismatch.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A help desk has three new tickets: the VP of sales, a long-time customer of the golf club next door, cannot change his desktop wallpaper; the order-entry system is down for all 80 call-center agents; one analyst's second monitor flickers. Which ticket should be worked first?

- A.** All three at the same level.
- B.** The VP's wallpaper, because of his rank.
- C.** The flickering monitor, since it came in first.
- D.** The order-entry outage.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A ticket affects only one user, but that user must submit a legal filing through the system within the hour. How do impact and urgency combine here?

- A.** Low priority, because only one user is affected.
- B.** Low impact but high urgency, so it may rank above higher-impact tickets that can wait.
- C.** High impact and low urgency.
- D.** Single-user tickets are always lowest priority.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A technician, ~~who just finished a certification course~~, gets a ticket that a branch office cannot print. She has asked users what changed and confirmed the printer shows offline. ~~The branch opened five years ago~~. What is her next step?

Target word: next step in the method **Decides:** The problem is identified; next is a theory of probable cause.

A	Absolute language	Most printer issues are solved locally.
B	Confusion set	Documentation is the final step.
C	Correct	Follows identifying the problem.
D	Inverted operation	Jumps to a fix before any theory or test.

2. Answer D Concept 1 · transfer

A technician fixed a switch port configuration, and the user's PC now connects. What should happen before closing the ticket?

Target word: steps after the fix **Decides:** Verify full function, then document.

A	Inverted operation	That step came earlier.
B	Adjacent category	Unneeded change with no link to the issue.
C	Dropped element	Drops verification and documentation.
D	Correct	The final steps of the method.

3. Answer A Concept 2

A user in the accounting department, ~~which recently moved to the third floor~~, says no websites load. The technician sees the laptop's network link light is off. What should the technician check first?

Target word: first check **Decides:** No link light means a physical-layer problem.

A	Correct	Start with the layer that shows the failure.
B	Inverted operation	Top-down guess that ignores the clear physical symptom.
C	Camouflage	The move detail does not decide the check.
D	Adjacent category	A higher layer; no link means nothing above can work.

4. Answer A Concept 3

Staff at a law firm, ~~which has a busy front desk~~, can reach the file server by its IP address but not by the name files.corp.local. Which tool best tests the likely cause?

Target word: tool choice **Decides:** IP works, name fails: name resolution.

A	Correct	Tests DNS name resolution directly.
B	Confusion set	Shows the path, not name resolution.
C	Adjacent category	Physical connectivity is fine since IP works.
D	Dropped element	Already known to work; tests the wrong thing.

5. Answer B Concept 3 · transfer

Users can reach a remote site's server by name some days but not others, and the team wants to see which router along the path stops responding. Which tool fits?

Target word: tool to find where the path breaks **Decides:** Need hop-by-hop path information.

A	Confusion set	Shows local settings only.
B	Correct	Lists each hop and where replies stop.
C	Inverted operation	Tests names, not the path.
D	Absolute language	ping tests only end-to-end reachability.

6. Answer A Concept 4

A laptop at a dental office, ~~recently upgraded to a new operating system,~~ shows IP address 169.254.12.7 and cannot reach anything. What is the most likely cause?

Target word: meaning of 169.254 **Decides:** APIPA address.

A	Correct	169.254 is self-assigned when DHCP fails.
B	Inverted operation	169.254 is not a normal assigned address.
C	Confusion set	DNS failure does not change the IP address.
D	Camouflage	The OS upgrade does not assign this range by design.

7. Answer B Concept 5

A design firm must copy a 2 GB file to a branch over a 100 Mbps link. ~~The link has 20 ms latency and was installed by a local carrier.~~ Ignoring overhead, about how long will the transfer take?

Target word: transfer time **Decides:** 2 GB = 16,000 megabits.

A	Dropped element	2,000 MB / 100 — drops the bytes-to-bits factor of 8.
B	Correct	16,000 Mb / 100 Mbps = 160 s.
C	Inverted operation	Divides the rate by the size instead.
D	Camouflage	That is latency, not transfer time.

8. Answer C Concept 5 · transfer

A 500 MB backup must finish in 40 seconds. Ignoring overhead, what minimum throughput is needed?

Target word: solve for rate **Decides:** 500 MB = 4,000 Mb.

A	Inverted operation	Multiplies instead of dividing by time.
B	Dropped element	500 / 40 — forgets to convert bytes to bits.
C	Correct	4,000 Mb / 40 s = 100 Mbps.
D	Wrong stem number	The time given, not a rate.

9. Answer C Concept 6

A monitoring dashboard shows a core switch at 65% CPU. The network team, ~~which uses a popular monitoring product,~~ is asked if this is a problem. What is needed to answer?

Target word: how to judge the reading **Decides:** Whether 65% is normal for this switch.

A	Absolute language	Some devices run near that level normally.
B	Adjacent category	Unrelated to judging load.
C	Correct	A reading means little without normal for comparison.
D	Camouflage	Tool version does not tell normal load.

10. Answer D Concept 7

One port on a switch in a warehouse near the loading dock shows CRC errors climbing every minute. The device on that port is a label printer ~~bought last year.~~ Users say printing is slow. What is the most likely cause?

Target word: cause of CRC errors **Decides:** CRC errors are physical-layer signs.

A	Confusion set	Name resolution does not corrupt frames.
B	Camouflage	Its age does not cause CRC errors.
C	Adjacent category	Routing affects paths, not frame checksums on one port.
D	Correct	Frame corruption comes from the physical link.

11. Answer D Concept 8

A help desk has three new tickets: the VP of sales, ~~a long-time customer of the golf club next door,~~ cannot change his desktop wallpaper; the order-entry system is down for all 80 call-center agents; one analyst's second monitor flickers. Which ticket should be worked first?

Target word: priority **Decides:** High impact (80 agents) and high urgency (orders stopped).

A	Absolute language	Impact and urgency differ.
B	Inverted operation	Rank does not set priority.
C	Adjacent category	Arrival order is not priority.
D	Correct	Largest impact and highest urgency.

12. Answer B Concept 8 · transfer

A ticket affects only one user, but that user must submit a legal filing through the system within the hour. How do impact and urgency combine here?

Target word: reason about the two factors **Decides:** Low impact, high urgency.

A	Dropped element	Drops urgency.
B	Correct	Priority weighs both factors.
C	Inverted operation	Swaps the two factors.
D	Absolute language	Urgency can raise them.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	5	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	5	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Troubleshooting method order	Identify the problem → establish a theory → test it → plan and implement the fix → verify full function → document.	Changing configurations before gathering information, or skipping verification and documentation.	A question about the next step in a repair.
2	Layered isolation	Work through the layers methodically — often bottom-up: link lights and cabling, then addressing, then routing, then application.	Reinstalling the application when the link light is off.	Symptoms that suggest a lower layer is down.
3	Choosing the right tool	ping tests reachability; traceroute shows the path and where it stops; nslookup/dig test name resolution; packet capture shows what is on the wire.	Using ping to diagnose a DNS failure, or traceroute to check a name.	“Works by IP but not by name” or “where along the path”.
4	APIPA addresses mean DHCP failure	A 169.254.x.x address means the host could not reach a DHCP server.	Treating 169.254 as a normal assigned address or a DNS issue.	A 169.254 address on a client.
5	Bandwidth, latency, and transfer time	Transfer time $\approx$ data size in bits / throughput in bits per second; latency is delay, not capacity.	Mixing bytes and bits, or treating low latency as high bandwidth.	File sizes in bytes and link rates in bits per second.
6	Baselines	Compare current metrics with a normal baseline to tell whether a reading is a problem.	Judging a number as high or low without knowing what is normal.	A metric reading with no context.
7	Interface errors point to the physical layer	Rising CRC errors, runts, or late collisions usually mean cabling, interference, or a duplex mismatch.	Blaming routing or DNS for interface error counters.	Error counters climbing on one interface.
8	Ticket priority from impact and urgency	Priority is set by combining impact (how many affected, how critical) and urgency (how fast it hurts), not by who asks or order received.	Working tickets first-come, first-served or by the requester’s rank.	Several tickets and a choice of which to work first.

Course scope (public description): Network Analytics and Troubleshooting teaches students to use network monitoring and analytics tools and practices that are common in the workplace in order to troubleshoot and fix complex computer networks. Students will follow a customer service model in identifying, classifying, investigating, and repairing network outages or problems. This course is designed as a hands-on experience where students will implement these techniques in a virtual space in order to produce a secure and functional deployed network.